

AE-590: Dual irradiation of [Ru(CO)(OH)2(PNP)] (AE-570)

Date: 2025-10-14

Tags: Dihydroxo PNP P(tBu)N(py)P(tBu)
[Ru(CO)(OH)2(PNP)] NMR AE Future 1H
31P Dual irradiation Power measurement
two-photon photocatalysis

Category: Two-photon

Status: Done

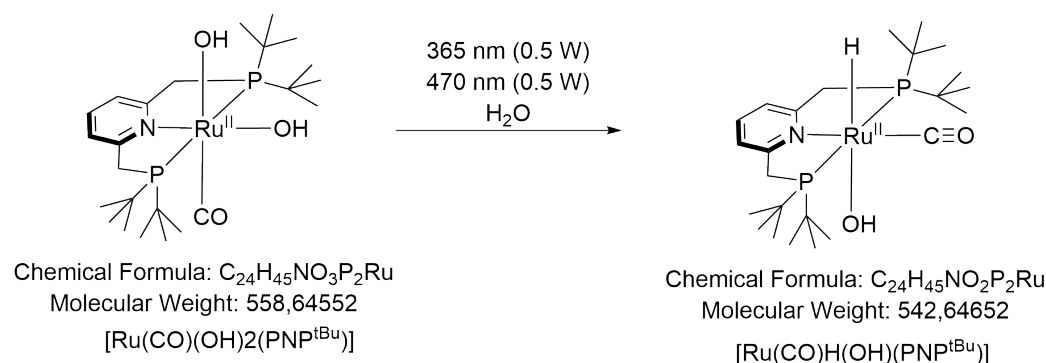
Created by: Alexander Eith

Objective

Testing dual irradiation effect of [Ru(CO)(OH)2(PNP)] at higher irradiance, to get better conversion after 16 h

Reaction scheme/sample structure

Radiation of Dihydroxo



ChemDraw File (linked): [AE-590.cdxml](#)

Literature/reference experiments

Literature	Procedure after https://doi.org/10.1039/D1EE01053K
Reproduction	/
Similar experiments	Two-photon - AE-519: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1) Two-photon - AE-511: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1) - reproduction II Two-photon - AE-502: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1) - reproduction Two-photon - AE-500: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1)

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [mmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Density (g/ml)	Volume [ml]	Concentration [mM]
[Ru(CO)(OH)2(PNP{tBu})]	Two-photon - AE-581: Stocksolution of [Ru(CO)(OH)2(PNP{tBu})] in water	/	/	/	/	/	558.65	/	3 * 0.6	concentrated (approx. 1.44 mM)

Irradiation Parameters

Power measured using [Power Meter - 843-R-USB + 919P-020-12](#) in [Equipment - Advanced power measurment setup V1.0](#)

	Name
Used Set-up	Equipment - Advanced irradiation setup V1.0 I
Irradiation set-up number	Equipment - Irradiation setup 2 (CEEC II, E004)

	Light Source Name	Used power source	Wavelength [nm]	Power Setting [mW]	Analogue Setting [-]
First light source	Light Source - UHP LED 365 nm-2	Power Sources - BLS-18000-1 1	365	500	9.68
Second light source	Light Source - UHP LED 470 nm	Power Sources - BLS-13000-1E	470	500	8.94

AE-590-1

Used beam combiner [Name or None]	Beam Combiner - Beam Combiner LCS-BC25-0409 -1
Used light source	365 + 470 nm
Irradiation distance [cm]	6.5
Thermostat temperature [°C]	25
Stirring speed [rpm]	/
Irradiation duration [h:min]	17:32

AE-590-2

Used beam combiner [Name or None]	Beam Combiner - Beam Combiner LCS-BC25-0409 -1
Used light source	365 nm
Irradiation distance [cm]	6.5
Thermostat temperature [°C]	25
Stirring speed [rpm]	/
Irradiation duration [h:min]	17:30

Procedure/observations

All steps (unless stated otherwise) were carried out under argon using either standard [Protocol - Schlenk Technique](#)

The synthesis/reactions were performed in the dark by either wrapping the reaction flasks in aluminum foil or turning the lab light and light in the fume hood off, when the flasks were uncovered.

Date	Time	Step	Observations	Pictures
13.10	16:05	Two Young type NMR tube was prepared according to Protocol - Preparation of NMR Sample (from step 3) using 0.65 mL of AE-581	AE-590-1 and AE-590-2	/
		From AE-590-1 an NMR was measured (AE-590-1-1)	AE-590-1-1 , slight yellow solution	AE-590-1-1.jpg
		From AE-590-2 an NMR was measured (AE-590-2-1)	AE-590-2-1 , slight yellow solution	AE-590-2-1.jpg
14.10	17:14	The power was adjusted using Power Meter - 843-R-USB + 919P-020-12 in Equipment - Advanced power measurement setup V1.0 (365 nm: 500 mW, 470 nm: 500 mW)	/	/
	17:19	AE-590-1 was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	no change	front before irradiation 1.jpg
	17:20	The irradiation of AE-590-1 was started	/	front during irradiation 1.jpg
15.10	10:52	The irradiation of AE-590-1 was stopped	darker orange	front after irradiation 1.jpg

	12:42	From AE-590-1 an NMR was measured (AE-590-1-2)	AE-590-1-2 , no change	AE-590-1-2.jpg
	14:20	One Young type NMR tube was prepared according to Protocol - Preparation of NMR Sample (from step 3) using 0.65 mL of AE-581 using a Prep work - AE-489: Capillary production	AE-590-3	
	14:50	From AE-590-3 an NMR was measured (AE-590-3-1)	AE-590-3-1 , slight yellow solution	AE-590-3-1.jpg
	16:41	AE-590-2 was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	no change	front before irradiation 2.jpg
	16:42	The irradiation of AE-590-2 was started	/	front during irradiation 2.jpg
16.10	10:12	The irradiation of AE-590-2 was stopped	no significant change	front after 2nd irradiation.jpg
	10:43	From AE-590-2 an NMR was measured (AE-590-2-2)	no change	AE-590-2-2.jpg

Analysis

Date	Time	Sample name	Analysis method	Analytical device	Solvent	Raw Data	Python script	Processed Data	Comparative Data	Interpretation
14.10	09:17	AE-590-1-1	¹ H, ³¹ P{ ¹ H} quant NMR	IAAC 400 MHz	H ₂ O	AE-590-1-1.zip	/	AE-590-1-1_10.nmr	Two-photon - AE-582: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-570-prod), activity test	As expected, before irradiation, reactant, and minor impurity
15.10	14:32	AE-590-1-2	¹ H, ³¹ P{ ¹ H} quant NMR	IAAC 400 MHz	H ₂ O	AE-590-1-2.zip	/	AE-590-1-1_10.nmr	Two-photon - AE-582: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-570-prod), activity test	As expected formation of hydride species and side products
14.10	13:47	AE-590-2-1	¹ H, ³¹ P{ ¹ H} quant NMR	IAAC 400 MHz	H ₂ O	AE-590-2-1.zip	/	AE-590-1-1_10.nmr	Two-photon - AE-582: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-570-prod), activity test	As expected, before irradiation, reactant, and minor impurity
16.10	12:17	AE-590-2-2	¹ H, ³¹ P{ ¹ H} quant NMR	IAAC 400 MHz	H ₂ O	AE-590-2-2.zip	/	AE-590-1-1_10.nmr	Two-photon - AE-582: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-570-prod), activity test	No conversion observed

Results

Sample	Yield (hydride species, cis) [%]	Yield (hydride species, trans) [%]	Yield (dihydroxo) [%]	Yield (main side product, 57 ppm) [%]	Yield (soluble side products) [%]	Yield (other side products) [%]	Data
AE-590-1	20	0	48	21	22	-9	AE-590-1_Yield_Determination.xlsx

AE-590-2: no evaluated, since LED was connected to wrong port of the beam combiner

AE-590-3: not measured, since a H₂O containing capillary was added not a D₂O containing one

Future recommendations

Old procedure	Problem	Suggested new procedure
Leaving aperture behind the the beam in the reactor	melted partially (molten aperture.jpg)	remove aperture for irrads with over 500 mW power
LEDs connected to the wrong ports of the beam combiner for irradiation with 365 nm	way less power output than desired	do connect them correctly
Use of capillary with H ₂ O and H ₃ PO ₄ for irradiation with 470 nm	no lock possible	make capillary with D ₂ O

Linked experiments

Prep work - [AE-489: Capillary production](#)

Two-photon - [AE-500: Dual Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\]](#) (AE-456-1)

Two-photon - [AE-502: Dual Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\]](#) (AE-456-1) - reproduction

Two-photon - [AE-511: Dual Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\]](#) (AE-456-1) - reproduction II

Two-photon - [AE-519: Dual Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\]](#) (AE-456-1)

Two-photon - [AE-581: Stock solution of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\]](#) in water

Two-photon - [AE-582: Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\]](#) (AE-570-prod), activity test

Linked resources

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Equipment - [Advanced irradiation setup V1.0 I](#)

Equipment - [Advanced power measurment setup V1.0 I](#)

Equipment - [Manual irradiation setup](#)

Equipment - [Irradiation setup 2 \(CEEC II, E004\)](#)

Light Source - [UHP LED 470 nm-1](#)

Light Source - [UHP LED 365 nm-2](#)

Power Meter - [843-R-USB + 919P-020-12](#)

Power Sources - [BLS-13000-1E](#)

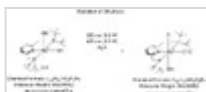
Power Sources - [BLS-18000-1 1](#)

Protocol - [Schlenk Technique](#)

Attached files

AE-590.png

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AE-590.cdxml

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AE-590-1_Yield_Determination.xlsx

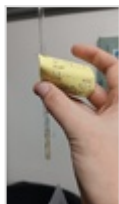
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AE-590-2-2.jpg

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front after 2nd irrad.jpg

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front during irrad 2.jpg

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AE-590-3-1.jpg

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molten aperture.jpg

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front after irrad 1.jpg

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front during irrad 1.jpg

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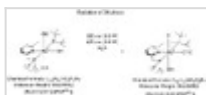


AE-590.cdxml

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AE-590.png

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Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=3182>